DESCENT OF FLATNESS ALONG A FINITE INJECTIVE MAP OF NOETHERIAN RINGS

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Allowed standard prerequisites (editorial).

- Flatness descends along faithfully flat ring extensions. Algebra, Lemma \ref{algebra-lemma-flatness-descends}, cited at proof lines 4647-4648. Justifies replacing the base ring by a finite locally free extension.
- Existence of a finite locally free extension making the target a quotient of a polynomial ring by monic polynomials. Lemma \ref{lemma-finite-split} and Lemma \ref{lemma-split-image}, cited at proof lines 4650-4653 and 4660-4661. Puts the finite injective map into the normal form that the rest of the proof uses.
- An integral ring map induces a surjective map on spectra. Algebra, Lemma \ref{algebra-lemma-integral-overring-surjective}, cited at proof lines 4662-4663. Shows that the ideals through which the module is flat have vanishing locus the whole spectrum.
- Flatness modulo an ideal whose vanishing locus is the spectrum, and lifting flatness modulo a nilpotent ideal. Lemma \ref{lemma-intersection-flat} and Algebra, Lemma \ref{algebra-lemma-lift-flatness}, cited at proof lines 4676 and 4678-4679. Combine the local flatness statements into flatness over the ring.

Source reference. Original lemma 0533: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6591–6591. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00HJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 645–645. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0531: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6589–6589. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0532: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6590–6590. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00GQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 616–616. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0522: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6555–6555. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 051J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6537–6537. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $R \to S$ be a finite injective homomorphism of Noetherian rings. Let M be an R-module. If $M \otimes_R S$ is a flat S-module, then M is a flat R-module.*

Proof. Let M be an R-module such that $M \otimes_R S$ is flat over S. By Algebra, Lemma 00HJ in order to prove that M is flat we may replace R by any faithfully flat ring extension. By Lemma 0531 we can find a finite locally free ring extension $R \subset R'$ such that $S' = S \otimes_R R' = R'[T_1, \ldots, T_n]/J$ for some ideal $J \subset R'[T_1, \ldots, T_n]$ which contains the elements of the form $P_i(T_i)$ with $P_i(T) = \prod_{j=1,\ldots,d_i}(T - \alpha_{ij})$ for some $\alpha_{ij} \in R'$. Note that R' is Noetherian and that $R' \subset S'$ is a finite extension of rings. Hence we may replace R by R' and assume that S has a presentation as in Lemma 0532. Note that $\mathrm{Spec}(S) \to \mathrm{Spec}(R)$ is surjective, see Algebra, Lemma 00GQ. Thus, using Lemma 0532 we conclude that $I = \bigcap J_{\underline{k}}$ is an ideal such that $V(I) = \mathrm{Spec}(R)$. This means that $I \subset \sqrt{(0)}$, and since R is Noetherian that I is nilpotent. The maps $\Phi_{\underline{k}}$ induce commutative diagrams

$$\begin{array}{ccc} S & \longrightarrow & R/J_{\underline{k}} \\ & \nwarrow \quad \nearrow & \\ & R & \end{array}$$

from which we conclude that $M/J_{\underline{k}}M$ is flat over $R/J_{\underline{k}}$. By Lemma 0522 we see that M/IM is flat over R/I. Finally, applying Algebra, Lemma 051J we conclude that M is flat over R. □